

CSE 311:

Data Communication

Instructor:
Dr. Md. Monirul Islam

Signals and systems

Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

Classification of Signals

Review

- Based on **continuity in time axis**
 - Continuous time
 - Discrete time
- Based on **continuity in amplitude axis**
 - Continuous amplitude (**analog**)
 - Discrete amplitude (**digital**)

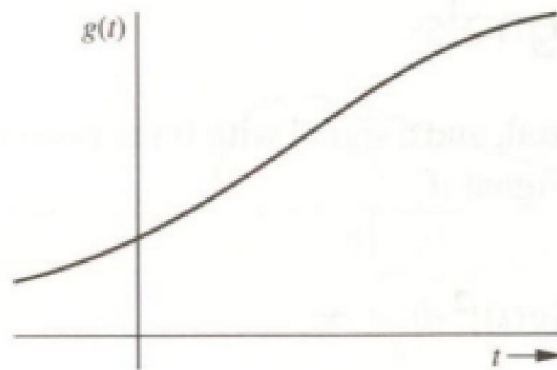
Classification of Signals

Review

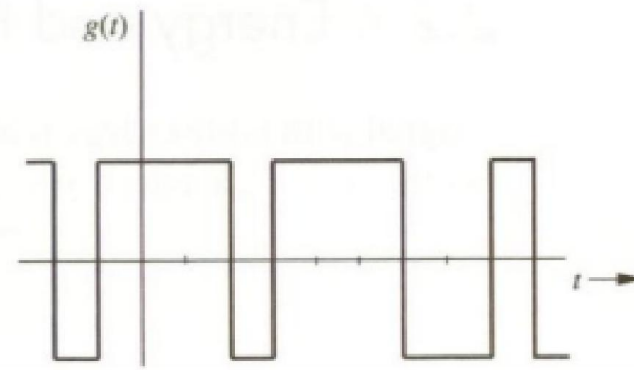
Continuous amplitude
(analog)

Discrete amplitude
(digital)

Continuous
time

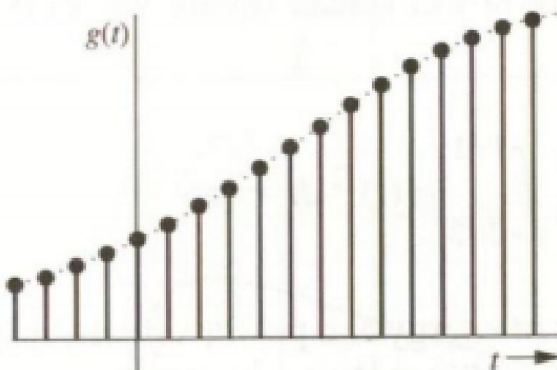


(a)

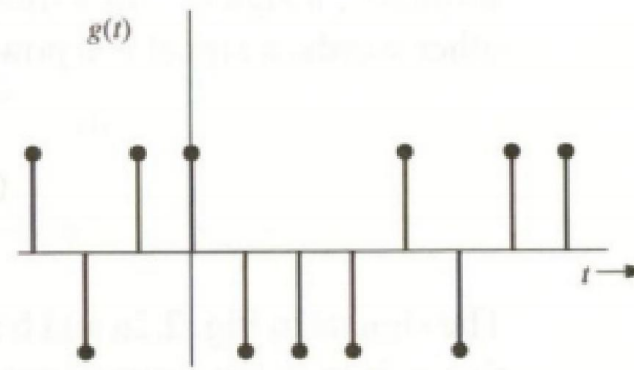


(b)

Discrete time



(c)



(d)

Periodic and Aperiodic signal

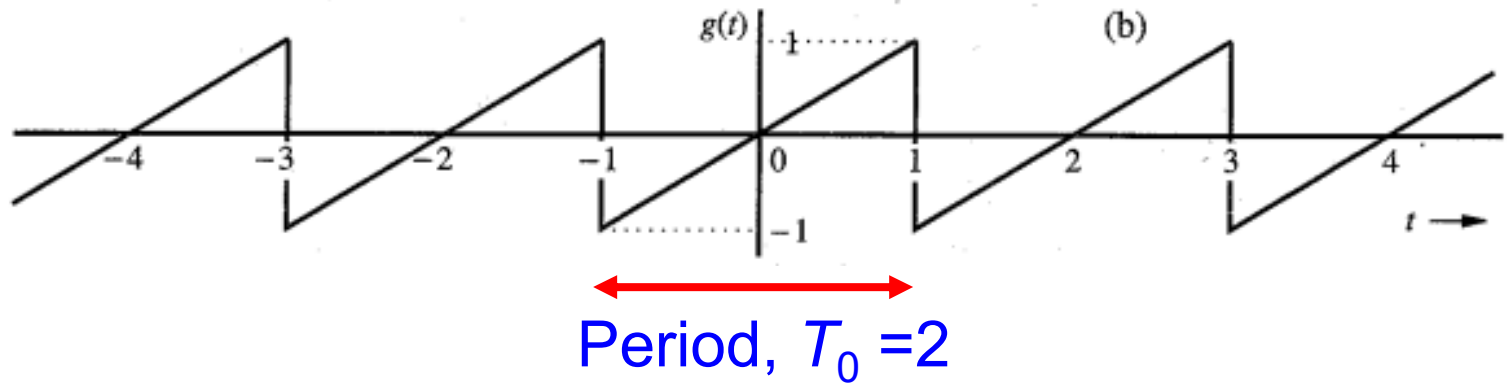
- Signal $g(t)$ is periodic for a positive constant T_0 so that

$$g(t) = g(t + T_0) \text{ for all } t$$

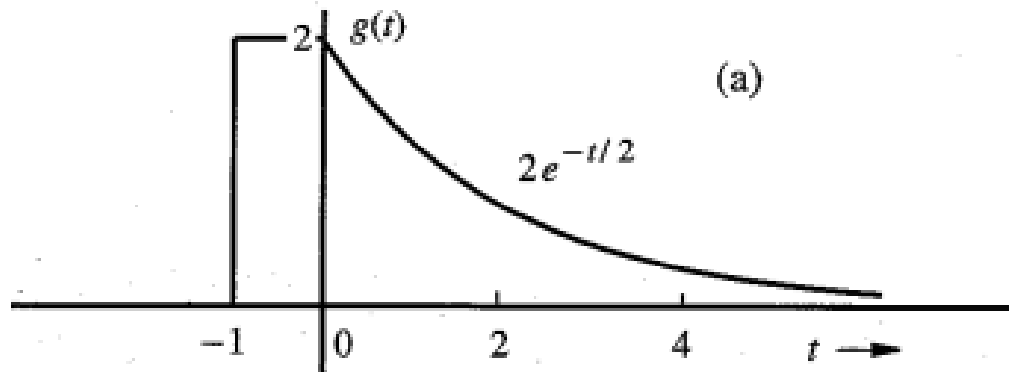
- Smallest T_0 is its period
- A signal is aperiodic if NOT periodic

Periodic and Aperiodic signal

Periodic
signal



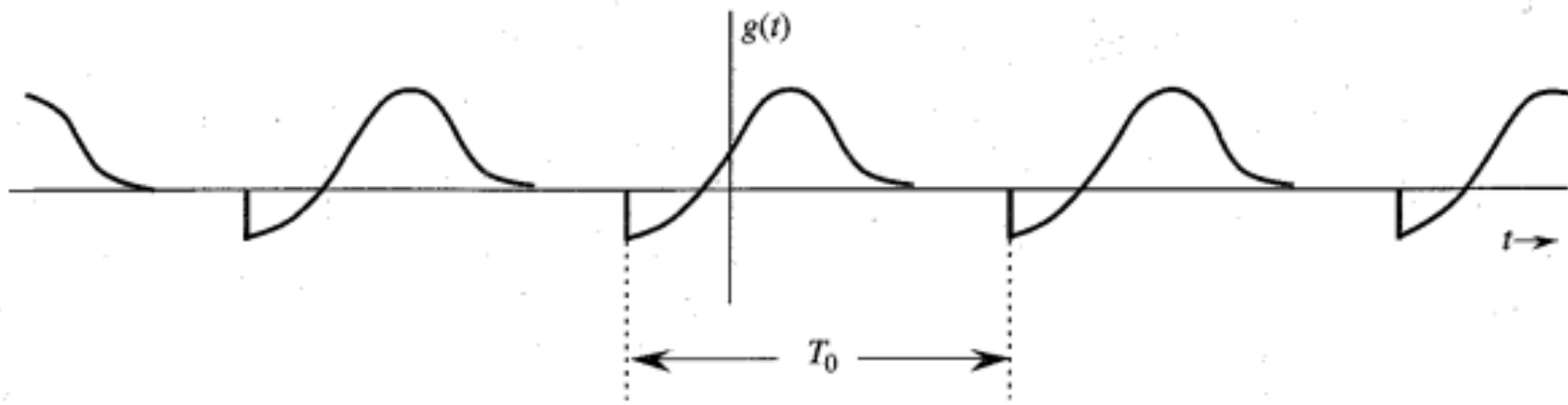
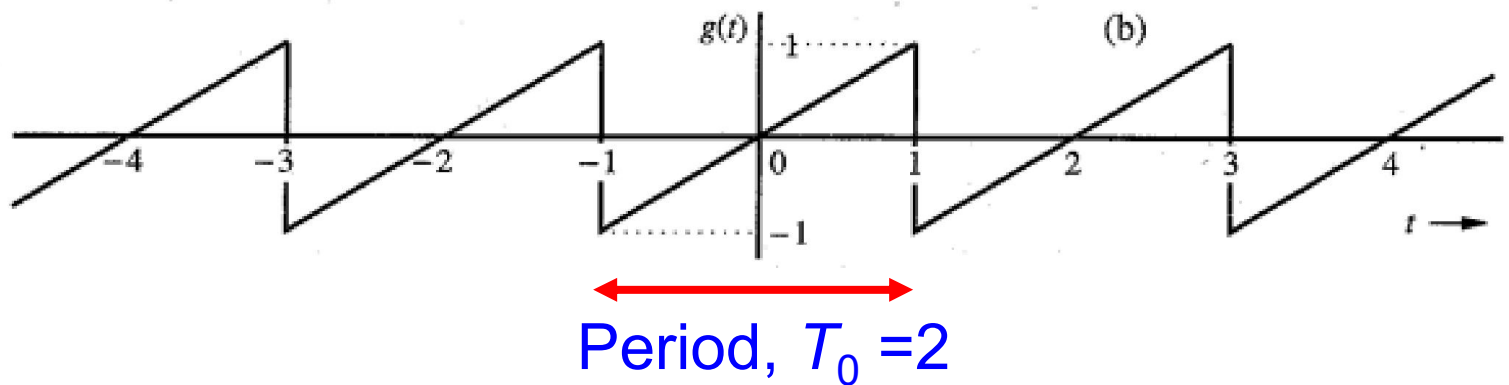
Aperiodic
signal



Periodic and Aperiodic signal

- Periodic signal starts at $t = -\alpha$ and continues forever

Periodic
signal

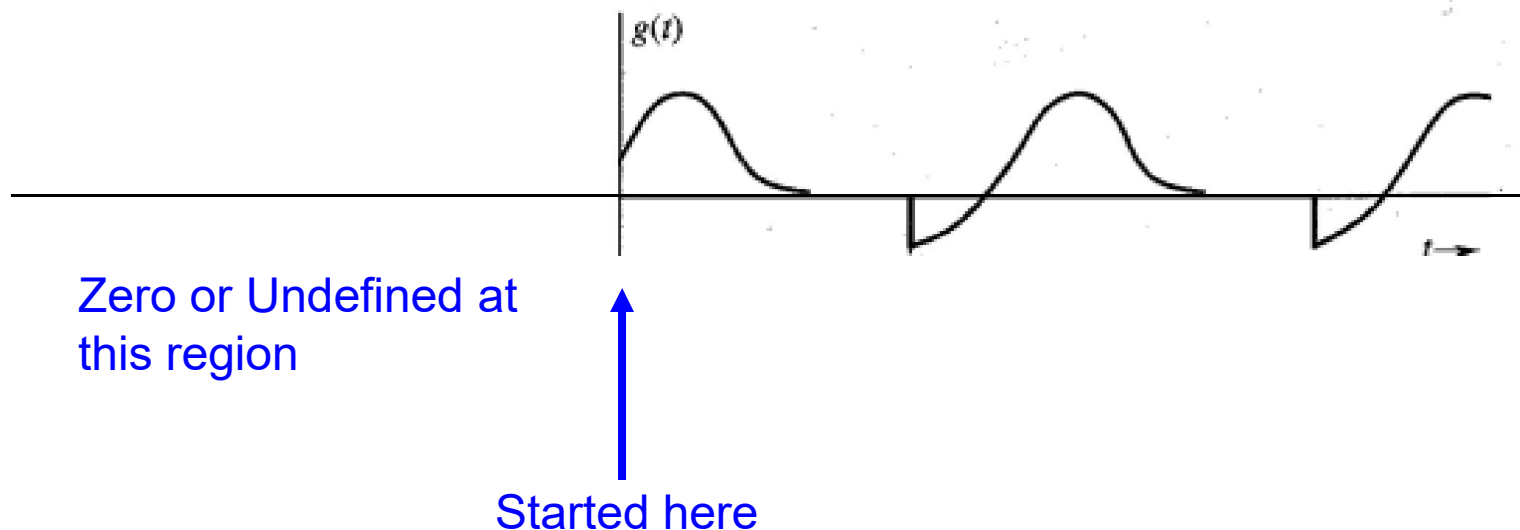


Periodic and Aperiodic signal

- Periodic signal starts at $t = -\alpha$ and continues forever
- cannot start at an finite time, say, $t = 0$, otherwise $g(t) = g(t + T_0)$ cannot be satisfied

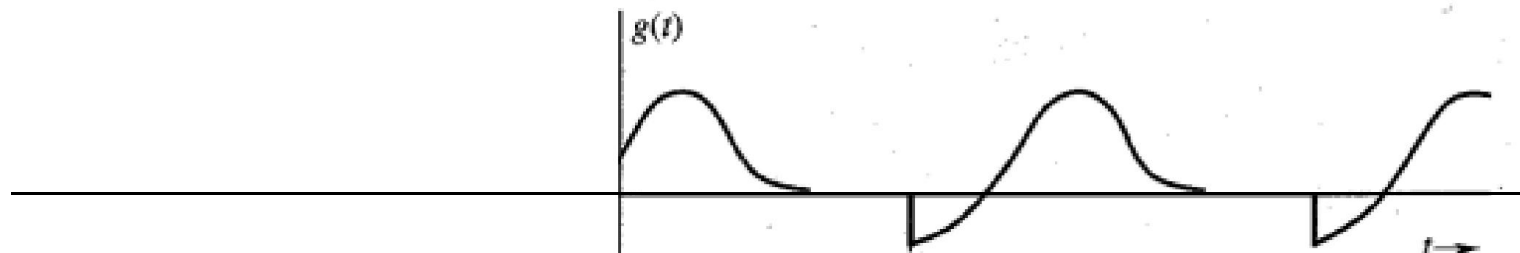
Periodic and Aperiodic signal

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Periodic and Aperiodic signal

- Periodic signal starts at $t = -\alpha$ and continues forever
- cannot start at an finite time, say, $t = 0$, otherwise $g(t) = g(t + T_0)$ cannot be satisfied



Zero or Undefined at
this region

At $t = -T_0$, $g(t) \neq g(t + T_0)$

Energy and Power signals

- Energy Signal: finite E_g

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt < \infty$$

- Power signal: finite P_g

$$0 < P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt < \infty$$

Energy and Power signals

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt < \infty$$

$$0 < P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt < \infty$$

- Finite E_g signal has zero P_g
- Finite P_g Signal has infinite E_g
- A signal **CANNOT** be both energy and power signal
- **Real life** signals are **energy** signals
- **Power signals** have **infinite duration**; impractical to generate
- **Periodic** signals are **power** signals

Deterministic and Random signals

- Deterministic
 - has complete physical description, mathematically or graphically
- Random
 - has only probabilistic description, e.g., mean value, rms, distribution
- All message signals are random

Signal Properties

Time shifting property

- Whatever happens in $g(t)$ at t second also happens in $\phi(t)$ T seconds later at instant $t + T$

$$\phi(t + T) = g(t)$$

Or,

$$\phi(t) = g(t - T)$$

Signal Properties

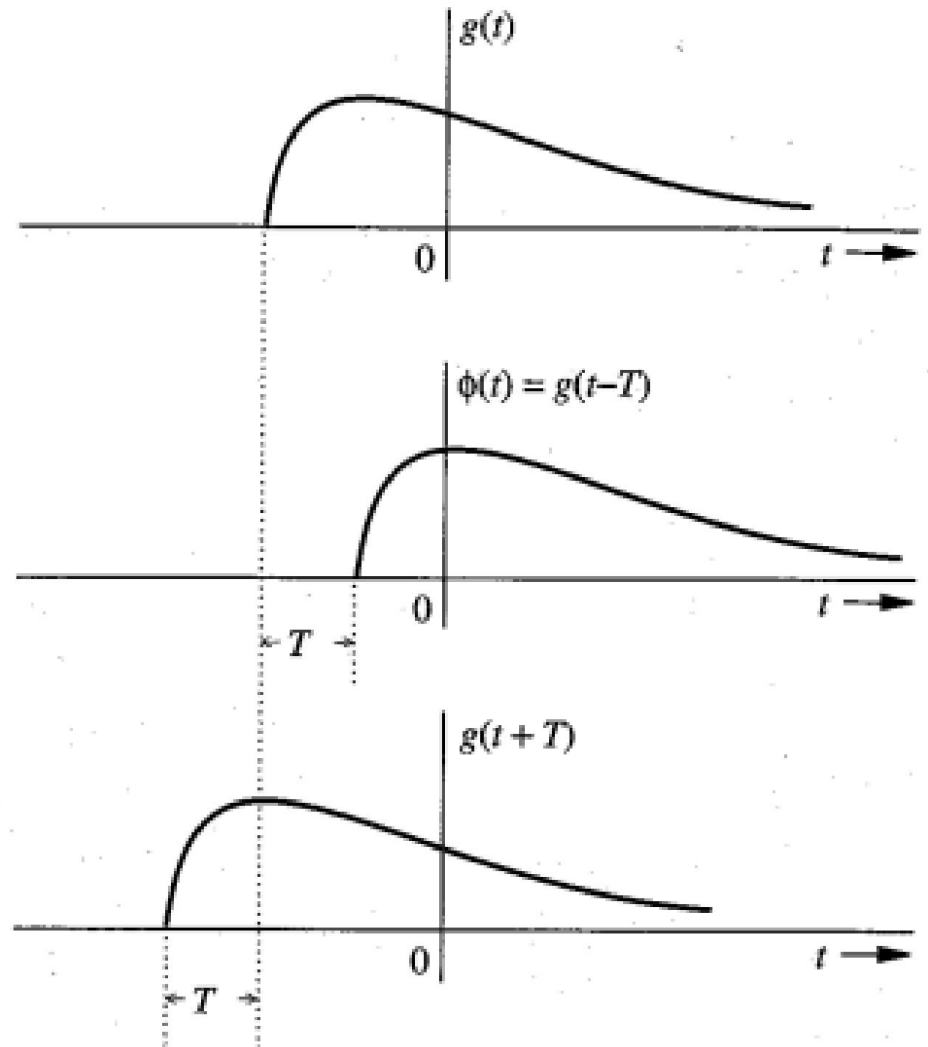
Time shifting property

$$\phi(t + T) = g(t)$$

$$\phi(t) = g(t - T)$$

Beginning T
seconds later

Beginning T
seconds earlier



Signal Properties

Time scaling property

- Compression or expansion
- Compression:
 - Whatever happens in $g(t)$ at t second also happens in $\phi(t)$ at t/a

$$\phi(t) = g(at), a > 1$$

Signal Properties

Time scaling property

- Compression or expansion
- Compression:
 - Whatever happens in $g(t)$ at t second also happens in $\phi(t)$ at t/a

$$\phi(t) = g(at), a > 1$$

- Expansion:
 - Whatever happens in $g(t)$ at t second also happens in $\phi(t)$ at at

$$\phi(t) = g\left(\frac{t}{a}\right), a > 1$$

Signal Properties

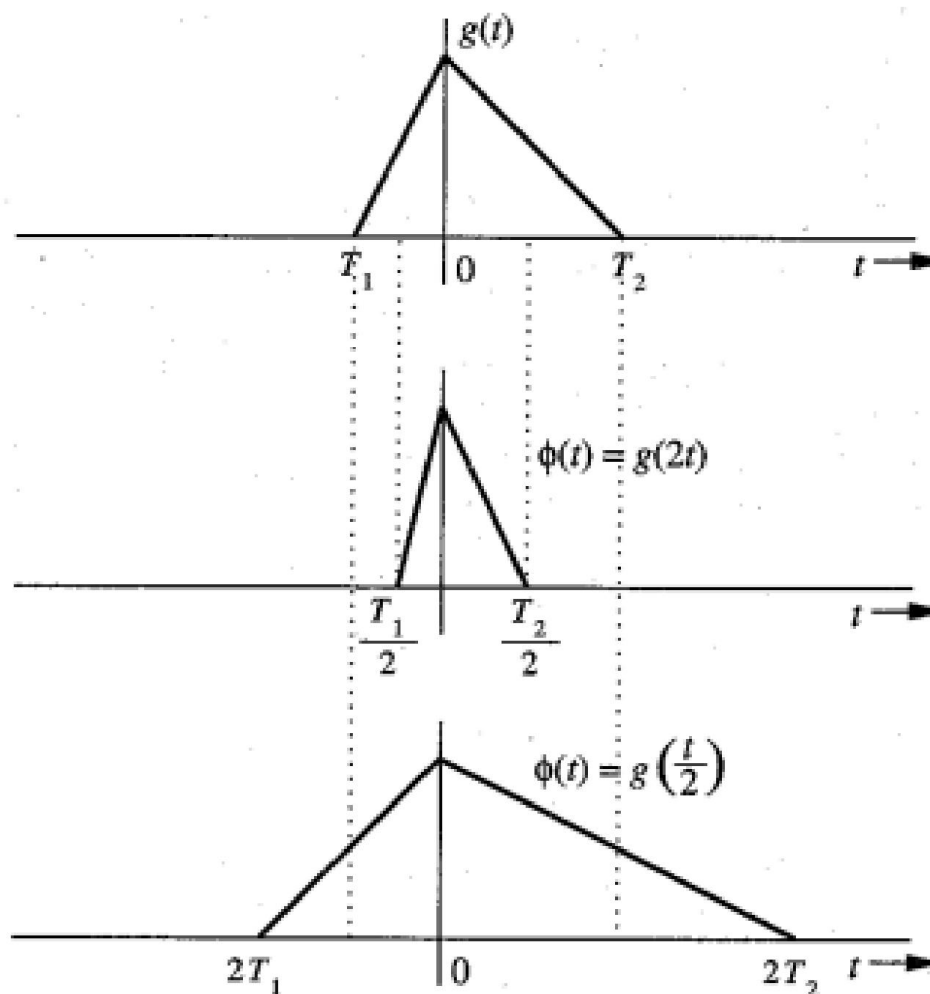
Time scaling property

$$\phi(t) = g(at)$$

$$\phi(t) = g\left(\frac{t}{a}\right)$$

Compression

Expansion



Signal Properties

Time inversion property

- Mirroring about vertical axis
- Whatever happens in $g(t)$ at t second also happens in $\phi(t)$ at $-t$
- Similar to time scaling where $a = -1$

$$\phi(-t) = g(t)$$

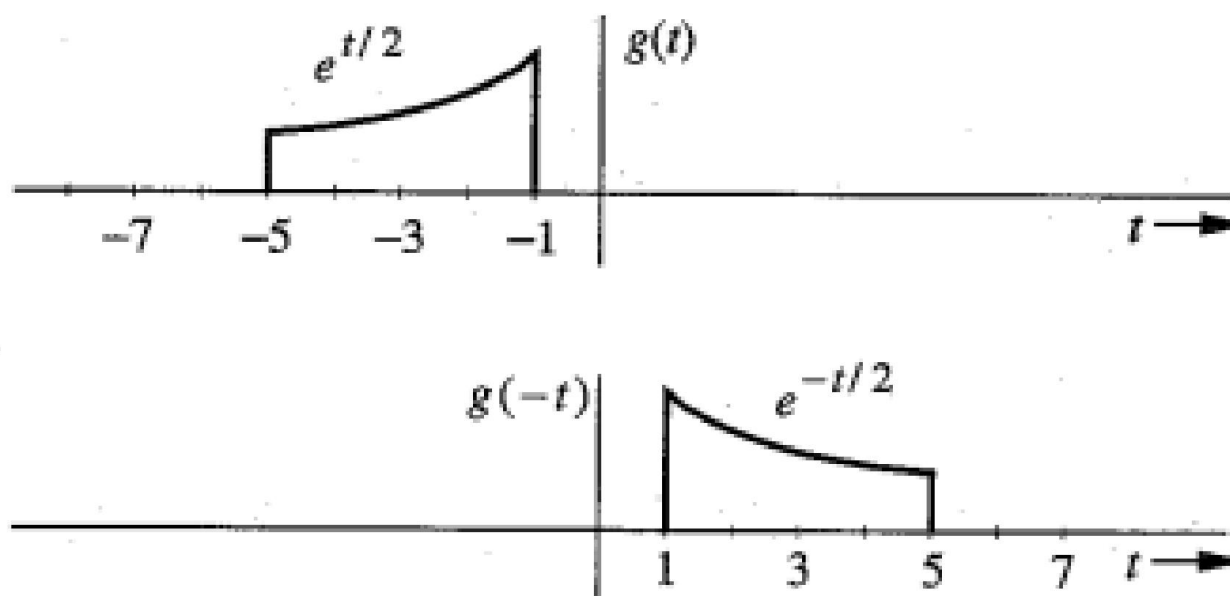
$$\phi(t) = g(-t)$$

Signal Properties

Example of Time inversion property

$$\phi(-t) = g(t)$$

$$\phi(t) = g(-t)$$



Unit Impulse Signal

- One of the most important signals, $\delta(t)$
- Also known as Dirac delta function

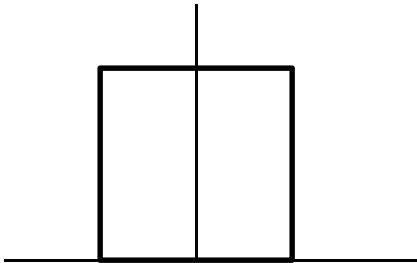
$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0 \end{cases}$$

with the constraint,

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

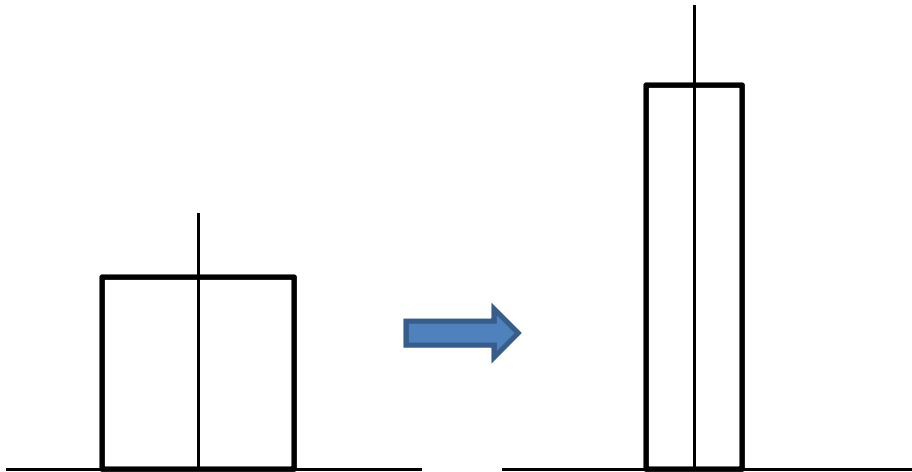
Unit Impulse Signal

A function with unit area under curve



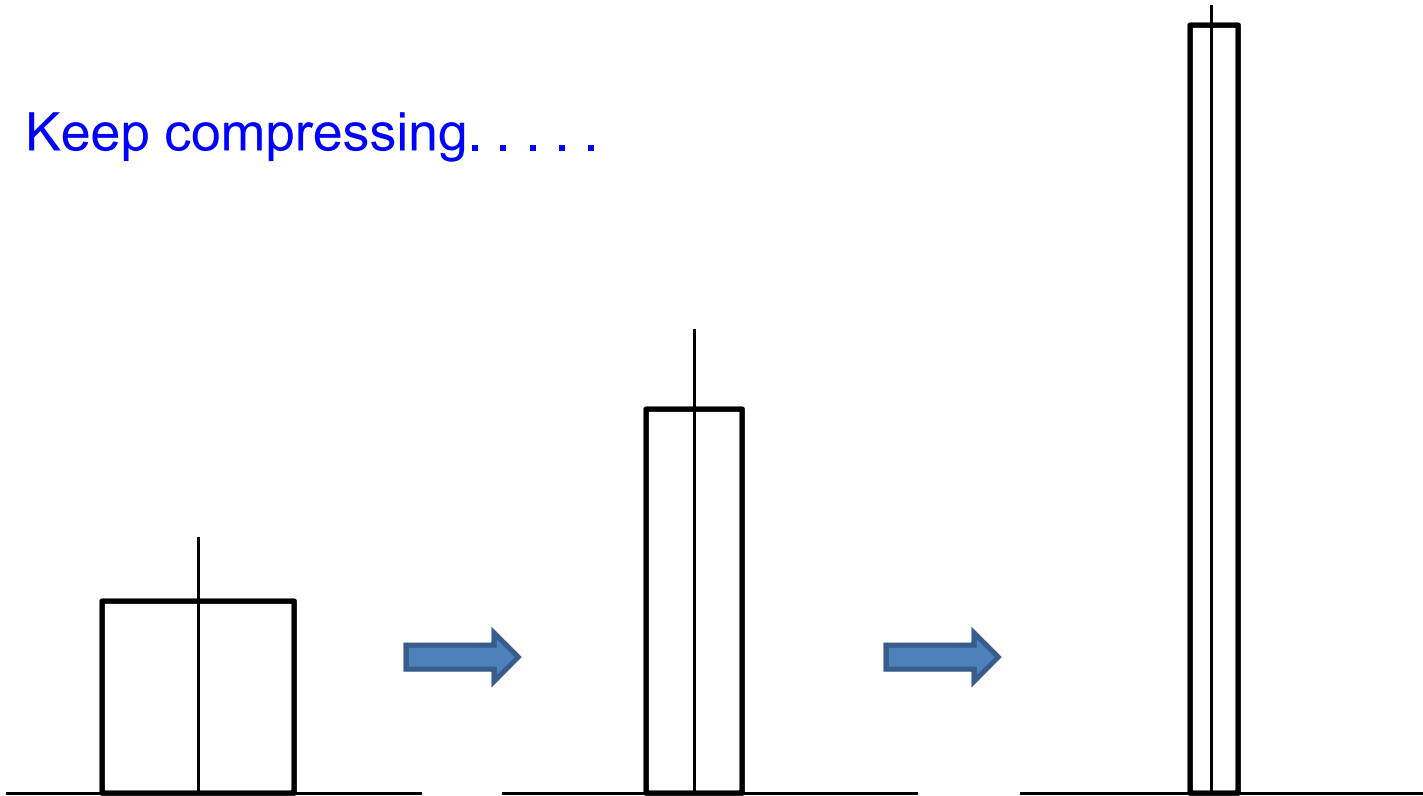
Unit Impulse Signal

Compress the function leaving
the area unchanged



Unit Impulse Signal

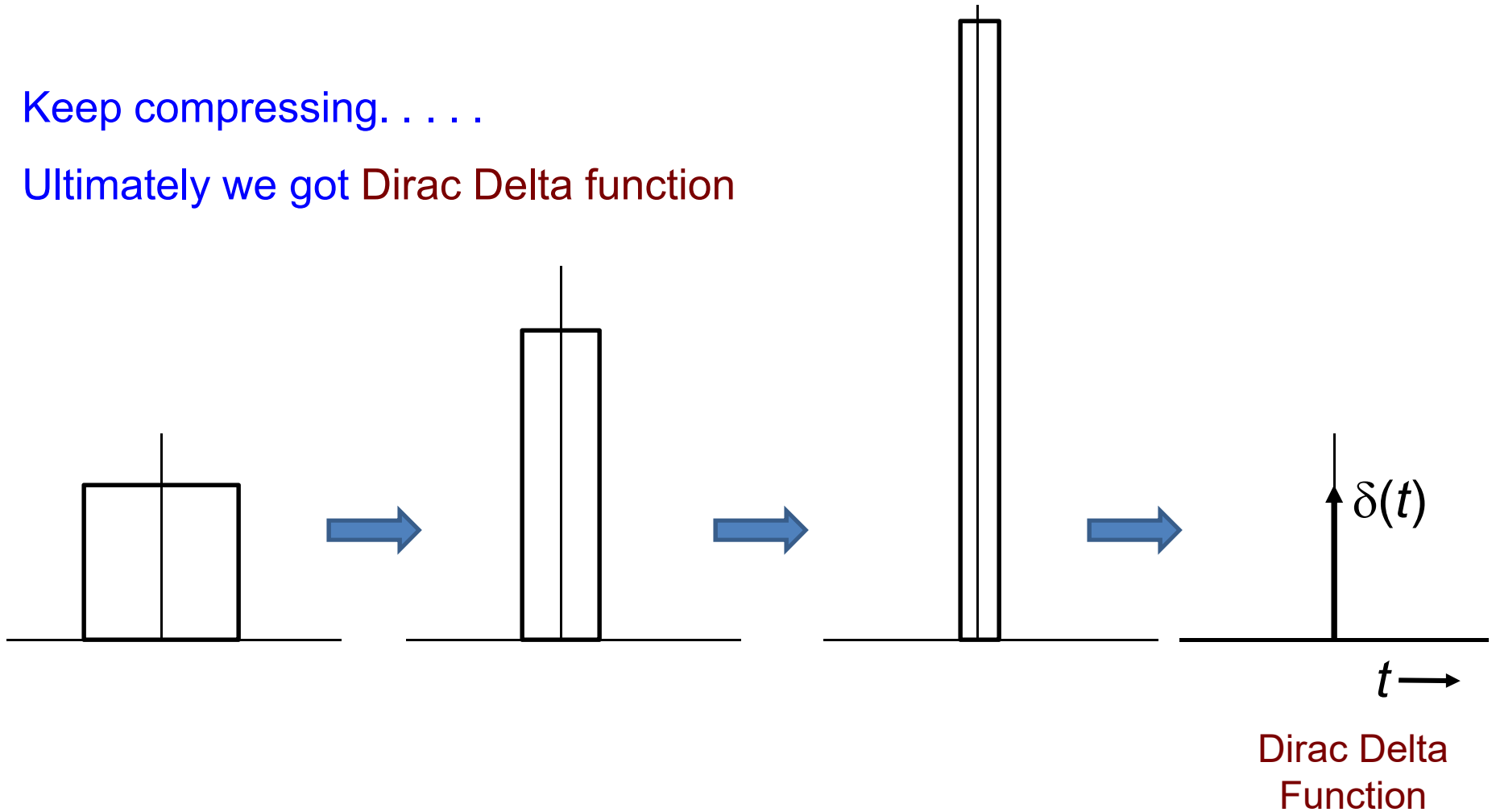
Keep compressing.



Unit Impulse Signal

Keep compressing.

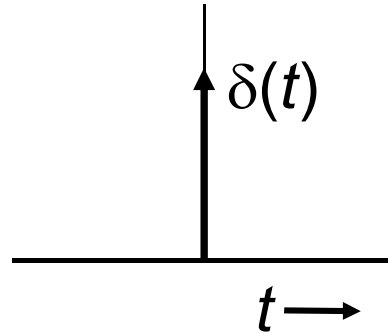
Ultimately we got Dirac Delta function



Unit Impulse Signal

$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0 \end{cases}$$

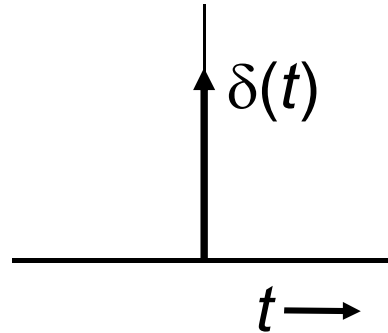
$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$



Unit Impulse Signal

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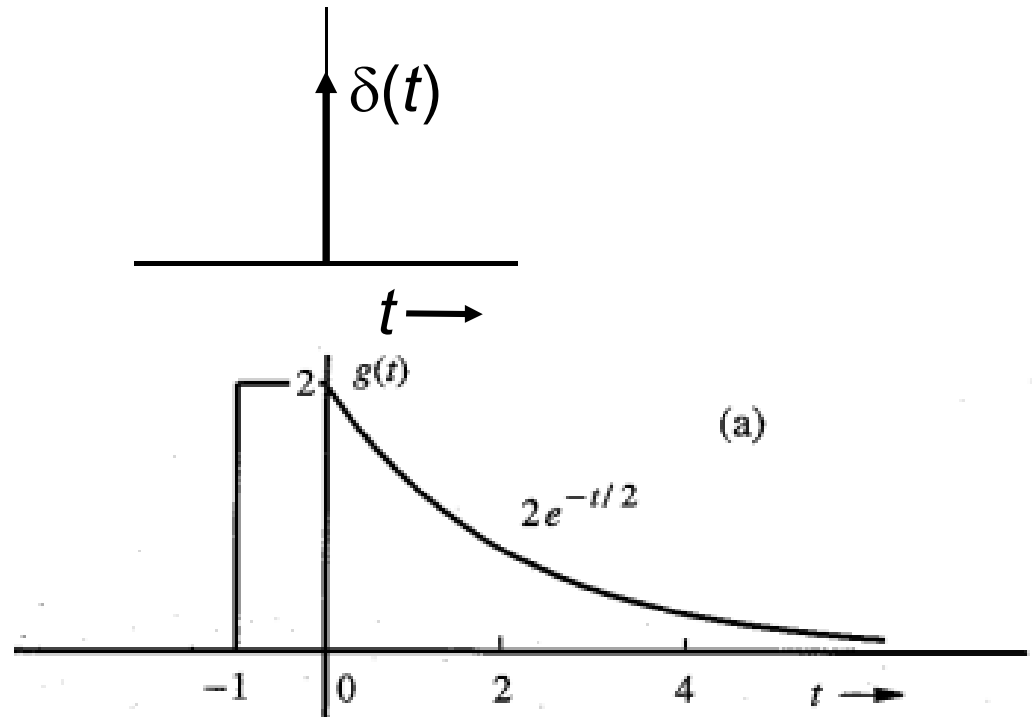
$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$



- Impulse location is at $t = 0$

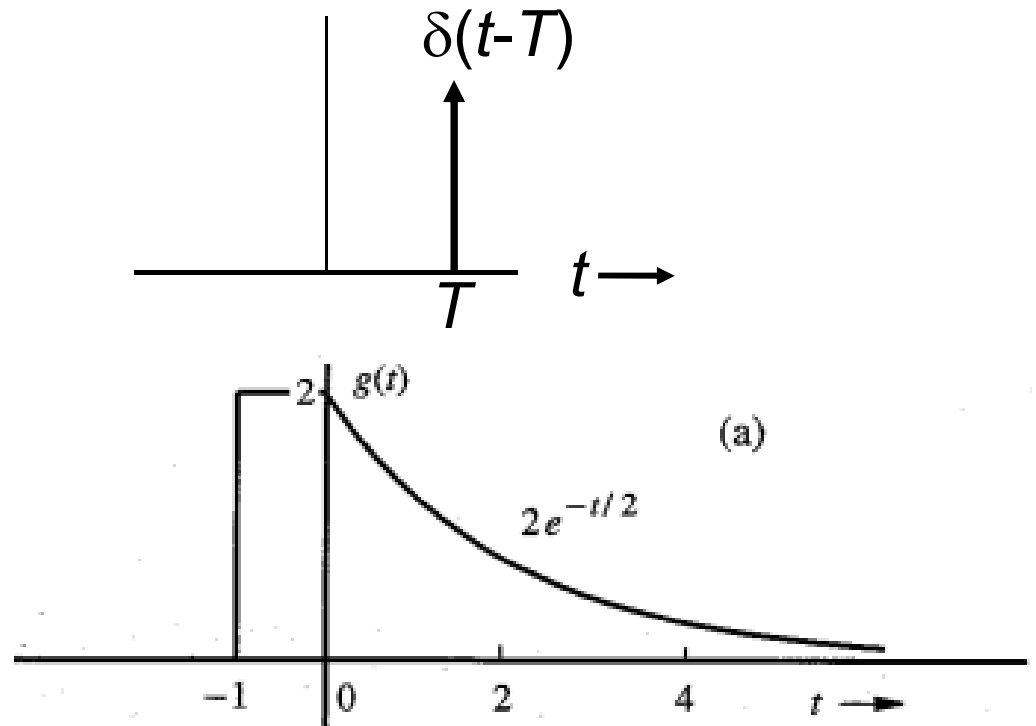
Multiplication of a Function by Impulse

$$\phi(t)\delta(t) = \phi(0)\delta(t)$$



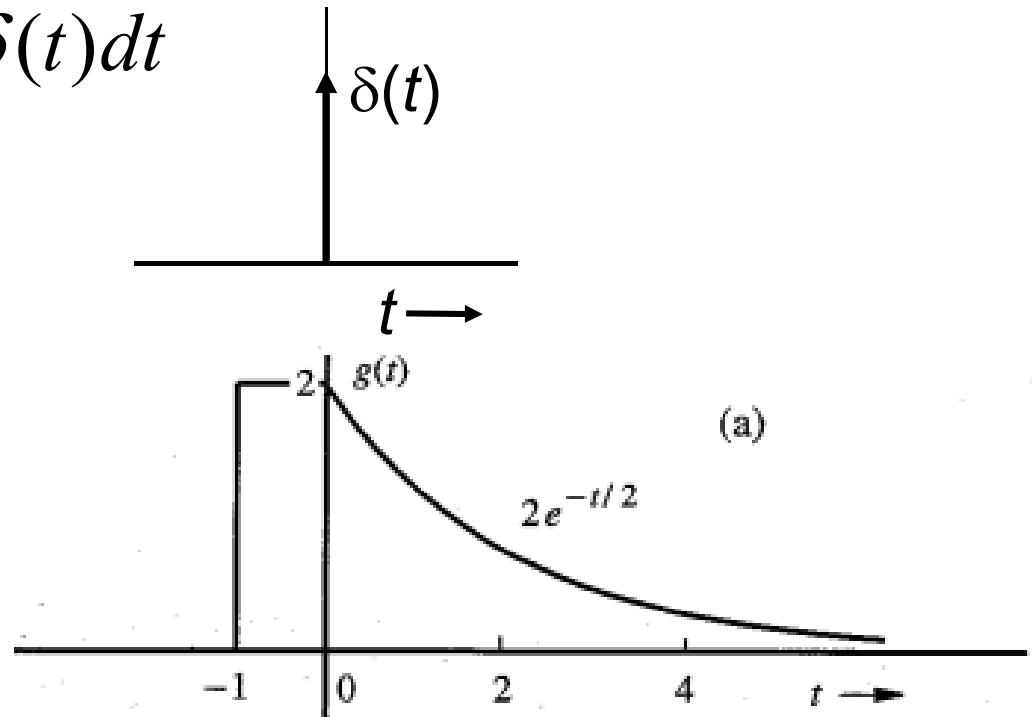
Multiplication of a Function by Impulse

$$\phi(t)\delta(t-T) = \phi(T)\delta(t-T)$$



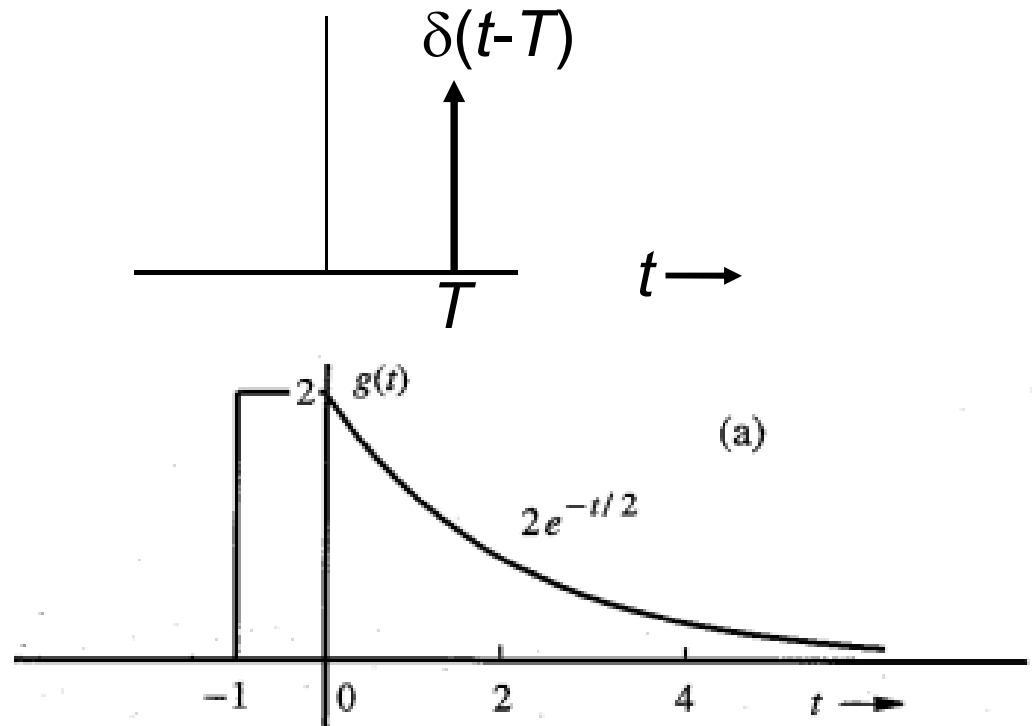
Sampling Property of the Unit Impulse Function

$$\int_{-\infty}^{\infty} \delta(t) \phi(t) dt = \phi(0) \int_{-\infty}^{\infty} \delta(t) dt$$
$$= \phi(0)$$



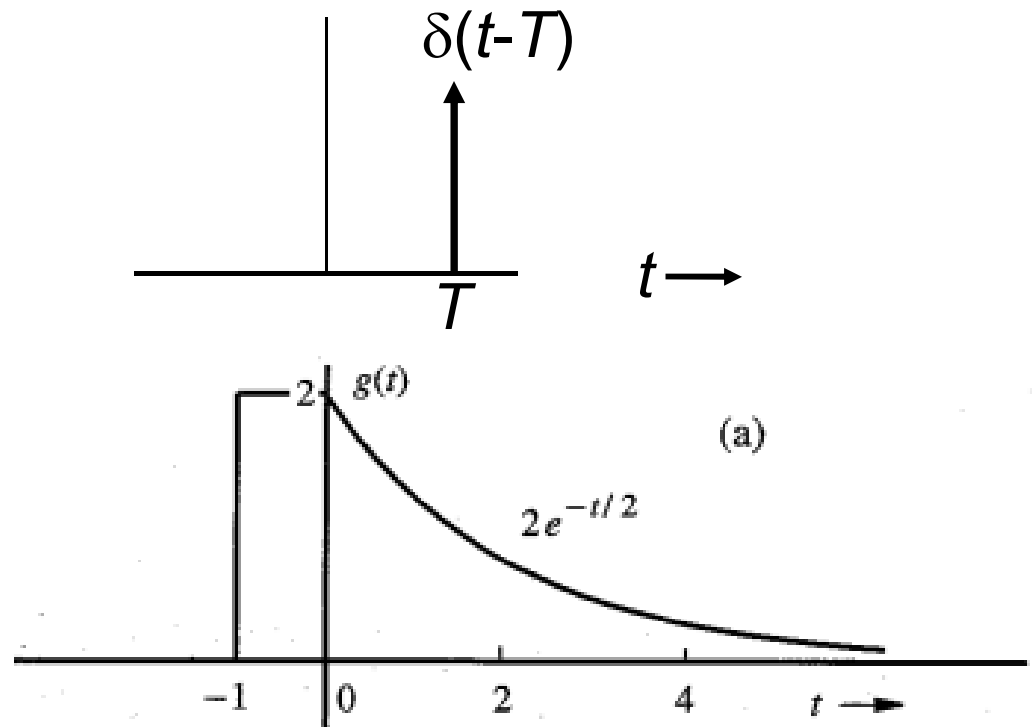
Sampling Property of the Unit Impulse Function

$$\begin{aligned} & \int_{-\infty}^{\infty} \delta(t-T)\phi(t)dt \\ &= \phi(T) \int_{-\infty}^{\infty} \delta(t-T)dt \\ &= \phi(T) \end{aligned}$$



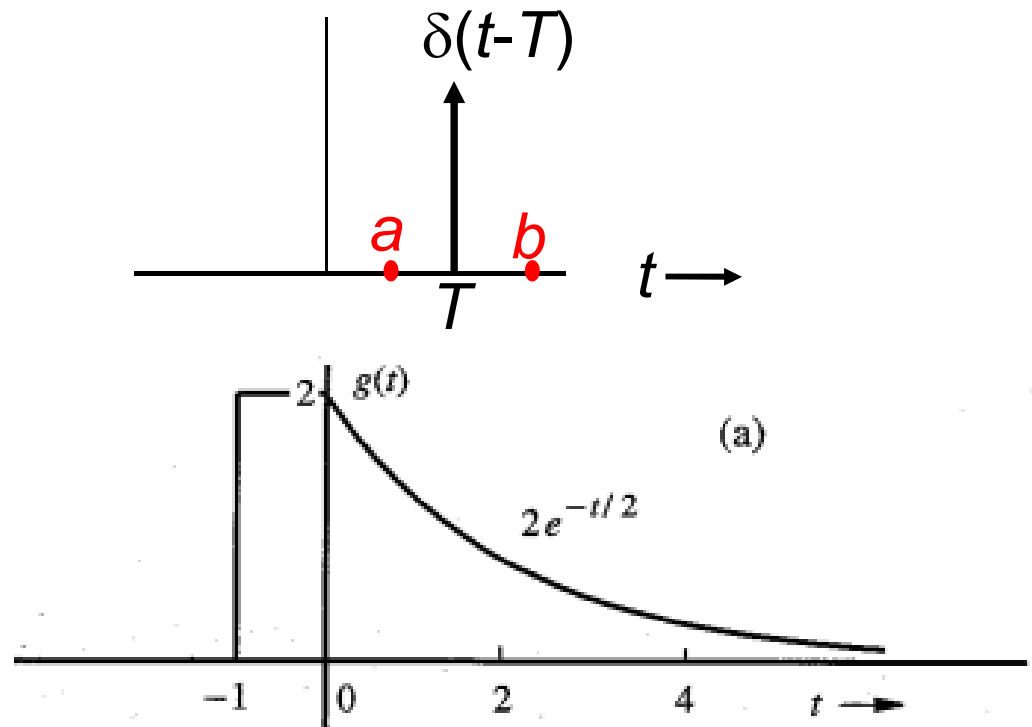
Sampling Property of the Unit Impulse Function

$$\int_a^b \delta(t-T)\phi(t)dt$$
$$= \phi(T) \int_a^b \delta(t-T)dt$$



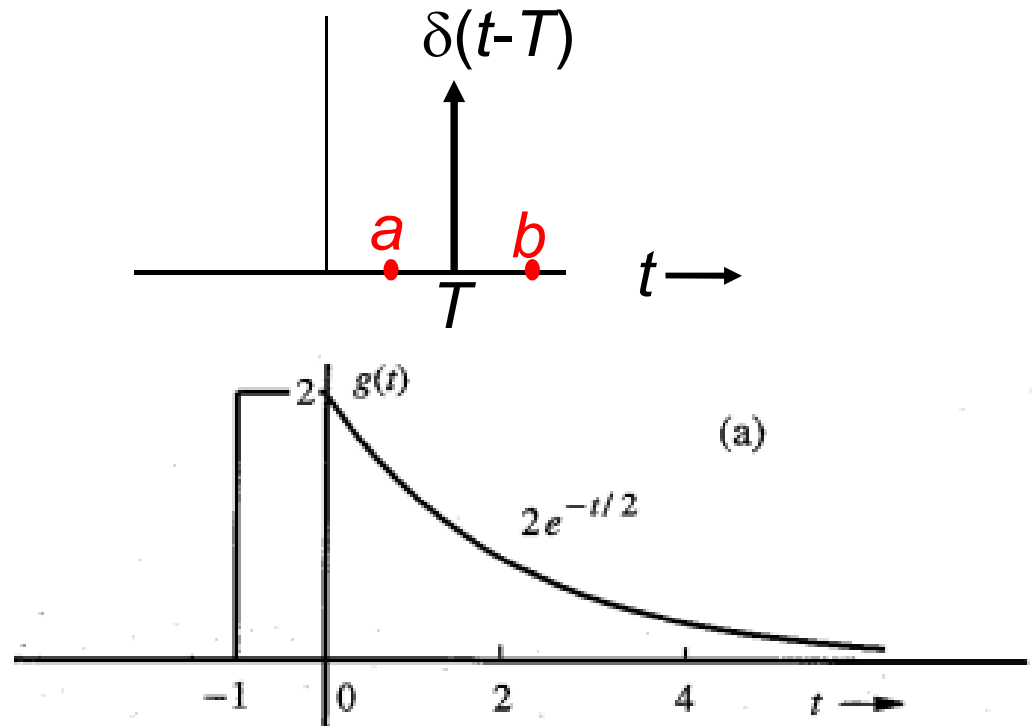
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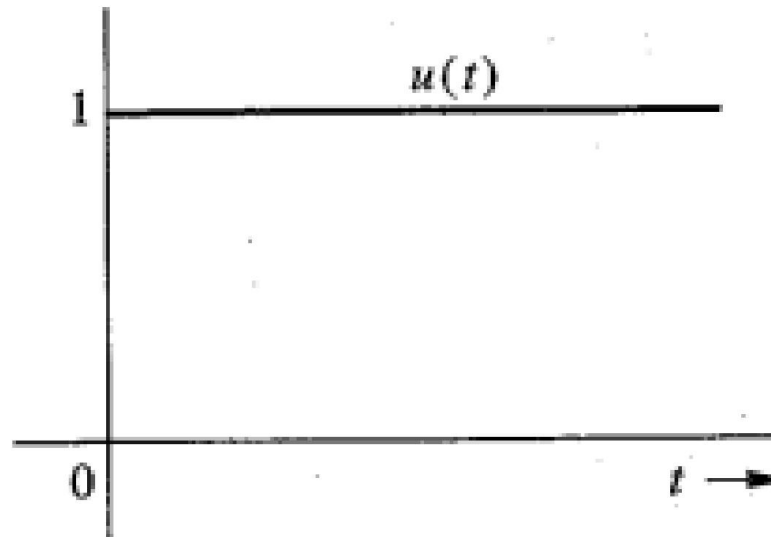


Sampling Property of the Unit Impulse Function

$$\begin{aligned} & \int_a^b \delta(t-T)\phi(t)dt \\ &= \phi(T) \int_a^b \delta(t-T)dt \\ &= \begin{cases} \phi(T) & a \leq T \leq b \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$



Unit step signal/function, $u(t)$



$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

Unit step signal/function, $u(t)$

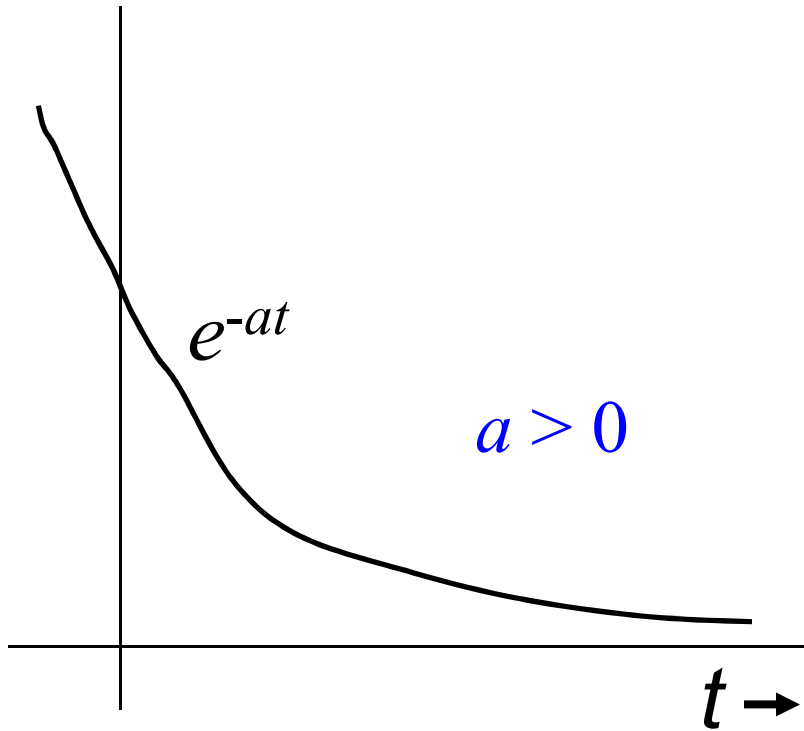
- Causal function, e.g., starts at $t = 0$

$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

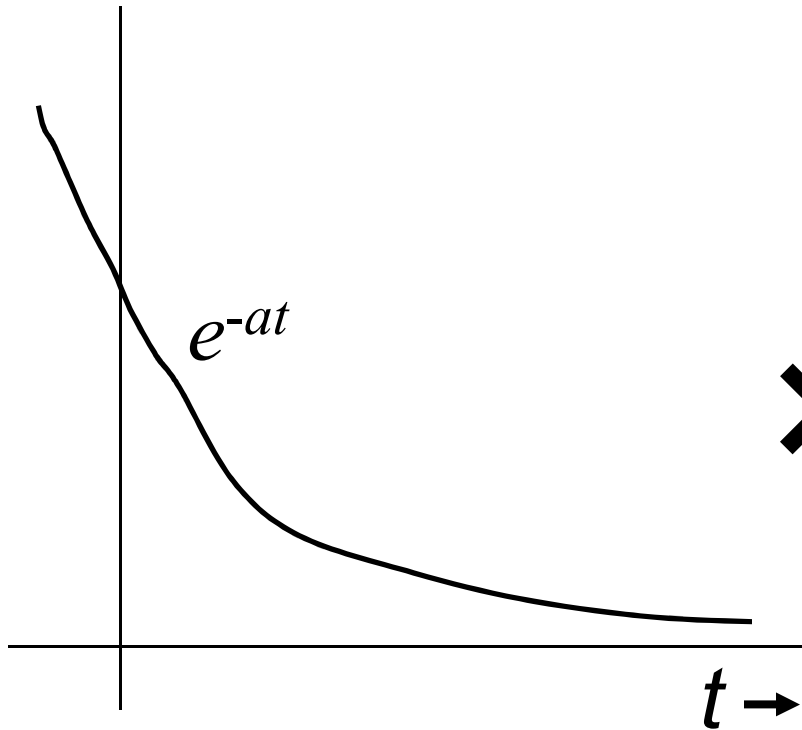
- $g(t)$ is causal if

$$g(t) = 0 \quad t < 0$$

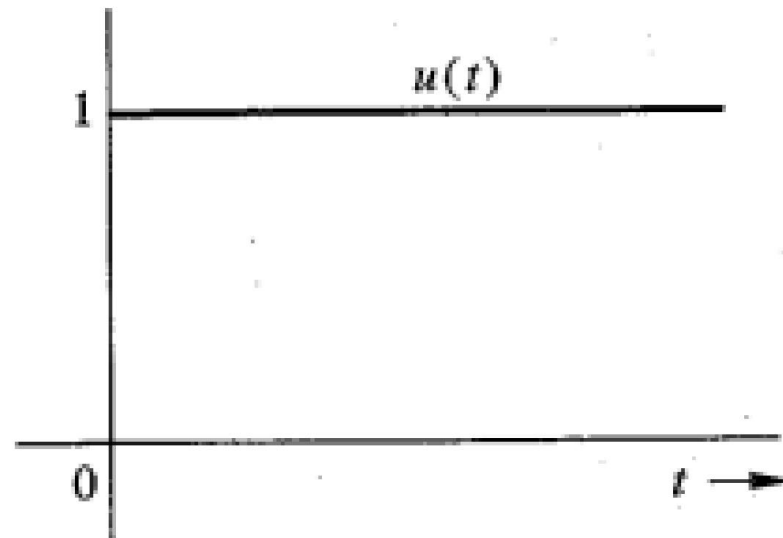
Use of $u(t)$ to start an everlasting function from $t = 0$



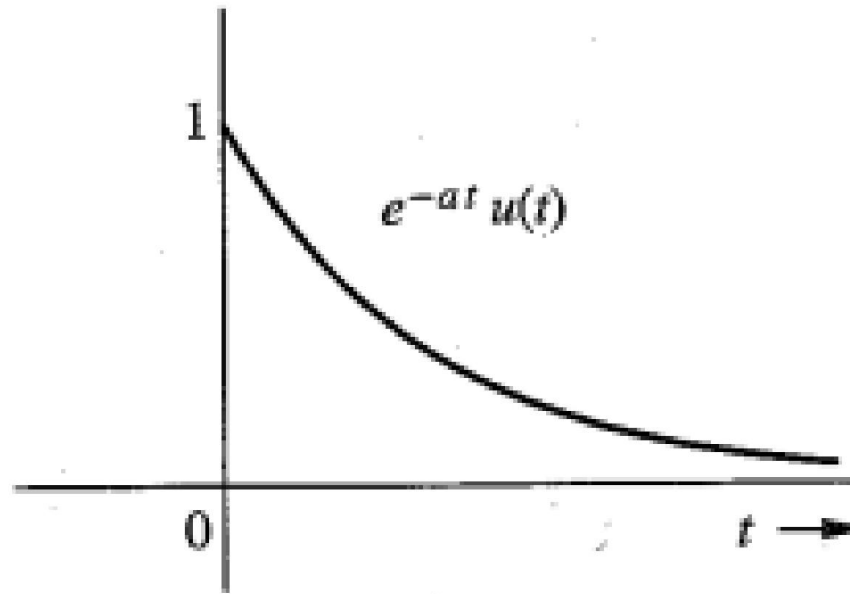
Use of $u(t)$ to start an everlasting function from $t = 0$



$\times$



Use of $u(t)$ to start an everlasting function from $t = 0$

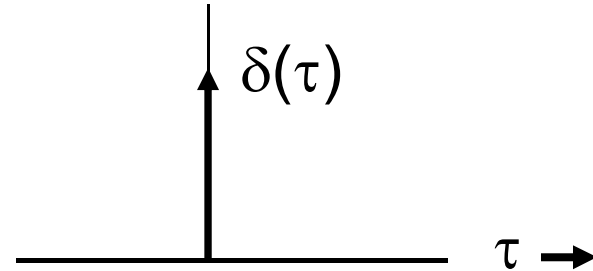


Alternate representation of $u(t)$ in terms of $\delta(t)$

$$\int_{-\infty}^t \delta(\tau) d\tau$$

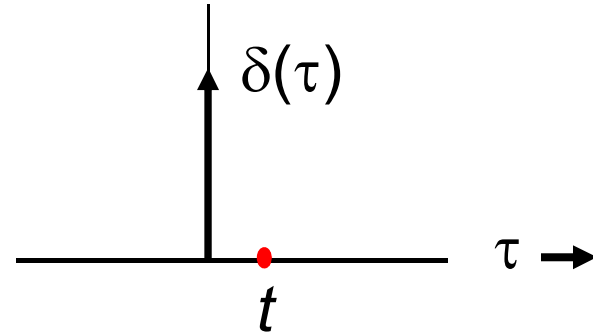
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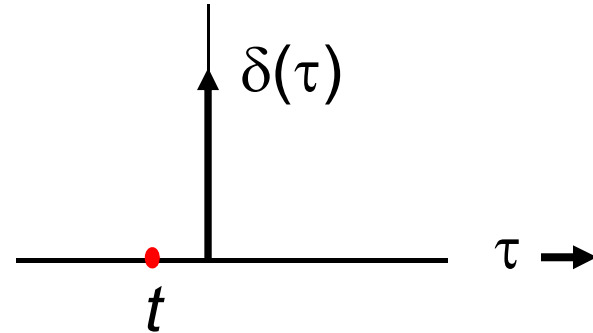
Alternate representation of $u(t)$ in terms of $\delta(t)$

$$\int_{-\infty}^t \delta(\tau) d\tau = 1$$



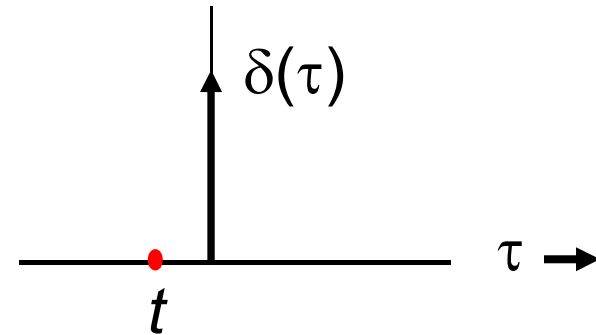
Alternate representation of $u(t)$ in terms of $\delta(t)$

$$\int_{-\infty}^t \delta(\tau) d\tau = 0$$



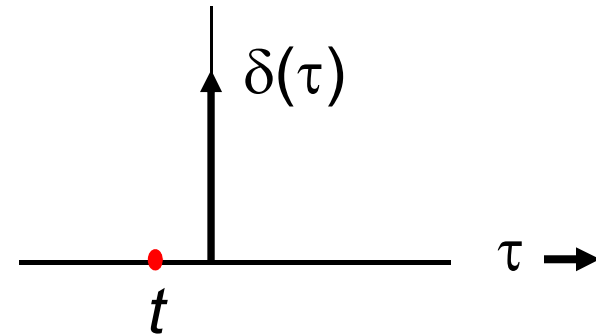
Alternate representation of $u(t)$ in terms of $\delta(t)$

$$\int_{-\infty}^t \delta(\tau) d\tau = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$



Alternate representation of $u(t)$ in terms of $\delta(t)$

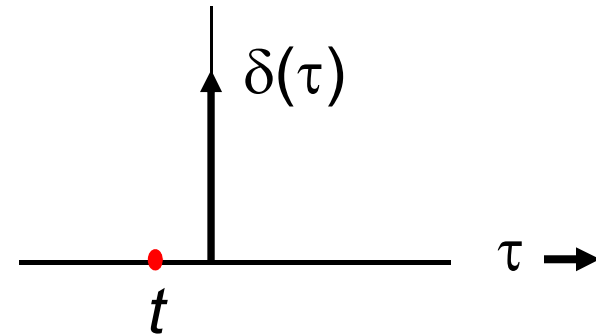
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$$u(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

Alternate representation of $u(t)$ in terms of $\delta(t)$

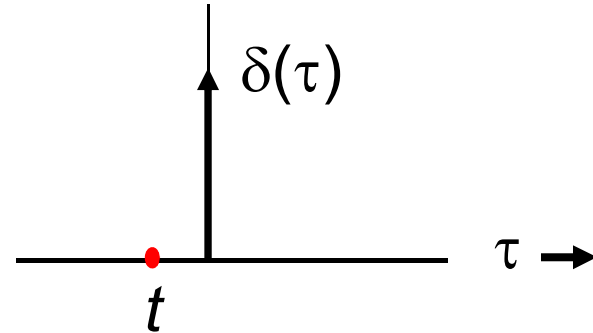
$$\int_{-\infty}^t \delta(\tau) d\tau = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases} \\ = u(t)$$



$$u(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

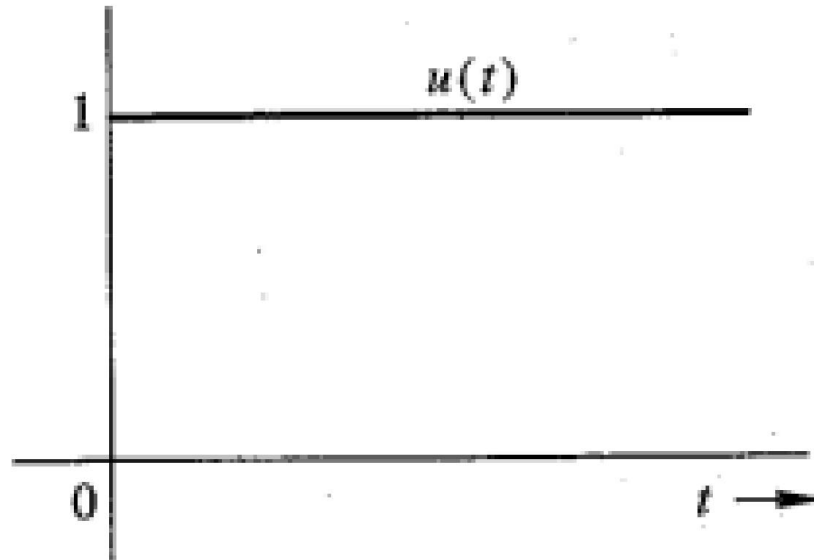
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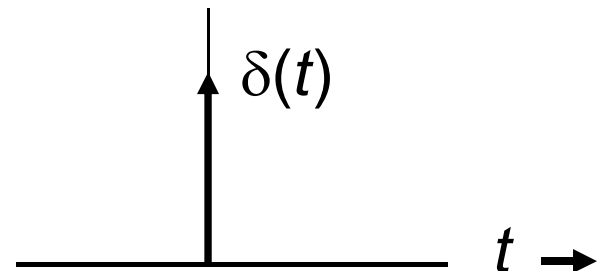


Alternate representation of $u(t)$ in terms of $\delta(t)$

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t)$$



$$\frac{du}{dt} = \delta(t)$$



Fourier Series of a periodic signal

‘Any function that *periodically repeats itself* can be expressed as the *sum of sines and/or cosines* of different frequencies’

- Joseph Fourier

Fourier Series of a periodic signal

‘Any function that *periodically repeats itself* can be expressed as the *sum of sines and/or cosines* of different frequencies’

- Joseph Fourier

$$g(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{jn\omega_0 t}$$

$$= \sum_{n=-\infty}^{n=\infty} D_n e^{jn2\pi f_0 t}$$

with, T_0 = period of $g(t)$
 f_0 = frequency of $g(t)$

$$\text{where, } D_n = \frac{1}{T_0} \int_{T_0} g(t) e^{-jn2\pi f_0 t}$$

Fourier Series

Fourier says:

Any function that *periodically repeats itself* can be expressed as the *sum of sines and/or cosines* of different frequencies

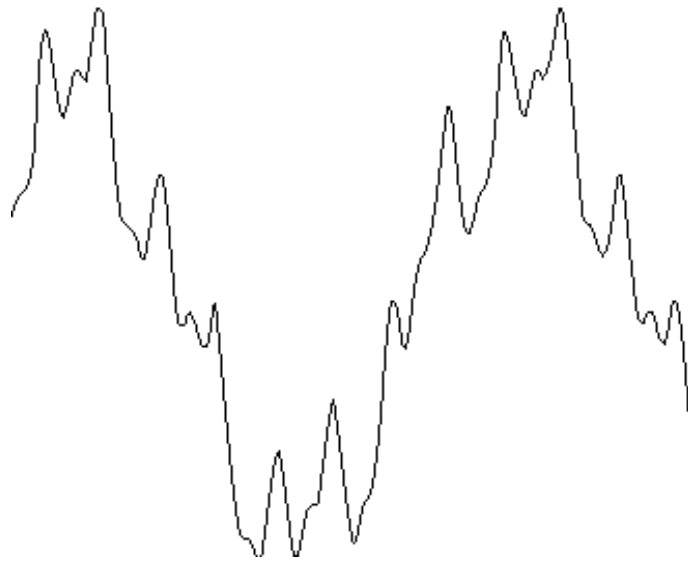
- Doesn't matter how complicated the function is!
- The summation is called *Fourier Series*

Fourier Series

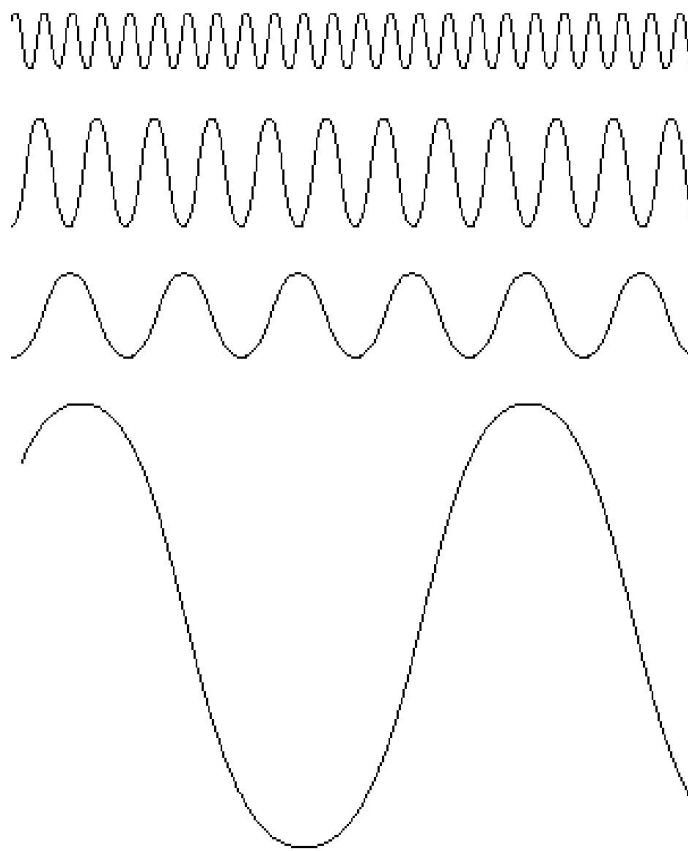


A complicated but periodic function

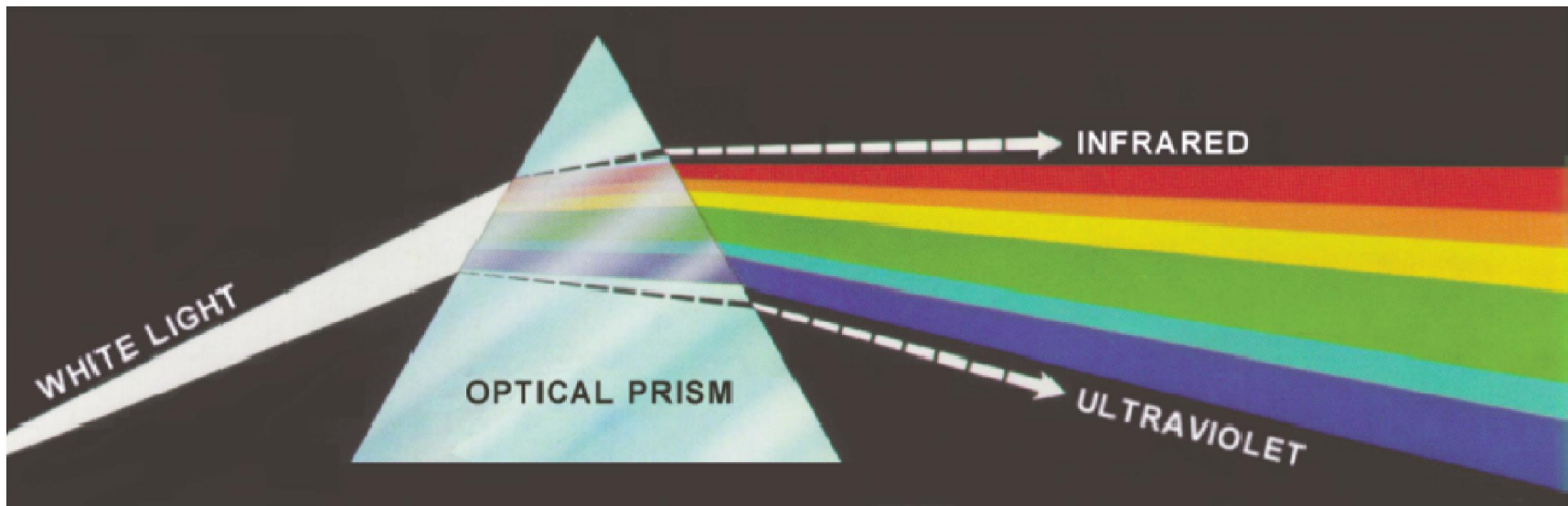
Complicated
periodic
function



Simpler
sine/cosine
functions



Fourier Series and Light Spectrum



A complicated
function/signal

Simpler
functions/signals